

Homework #1

Please **complete the following coding exercises**. Each exercise should be completed in a **separate Python script** file unless otherwise noted.

For this assignment, you will turn in a **zip folder** containing:

- **5 – Python (.py) code files** containing commented code for Exercises #1-5.

As a reminder, be sure to add the required comments at the top of **each** file (as shown below). You will not lose points for it on this assignment, but **in future assignments points will be deducted if omitted**.

```
# [Your Name]
# ECET 114 / ITC 150
# [Date]
# Explanation of the program goes here
```

Exercise #1: Hello World (Completed Together in Class)

For Exercise #1, you will write a program that outputs,

“Hello World! My name is [your name]!”

to the terminal, replacing [your name] with your own name.

```
Hello World! My name is Harrison Niswander!
```

Exercise #1: Example Output

Exercise #2: Name and Age (Completed Together in Class)

For Exercise #2, you will write a program that asks the user to input their name and age. Using the data inputted from the user, you will print out a message using their name and age.

```
Please enter your name: Harrison  
Please enter your age: 22  
Hello, Harrison ! You are 22 years old.
```

*Exercise #2: Example Output***Exercise #3: Candy Shop Manager**

For Exercise #3, you will write a candy shop management program that asks a user to enter an order for a candy shop, then lets the user know that the order is being processed.

```
Welcome to Willy Wonka's Chocolate Factory!  
What would you like? Jaw Breaker  
Your order of Jaw Breaker is being prepared. Please wait a moment.
```

*Exercise #3: Example Output***Exercise #4: Cookie to Miles Conversion**

For Exercise #4, you will write a program that converts the number of cookies inputted by the user into how many miles long it is using the following formula:

$$\text{Miles} = \left(\frac{\text{Cookies}}{1.23} \right) + 15 \cdot 1.1$$

Note: the miles calculated do not need to be formatted to a certain decimal length.

```
Please enter in a quantity of cookies: 1892  
Your inputed quantity of cookies is equal to 1554.7113821138212 miles
```

Exercise #4: Example Output

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Exercise #5: Formatting Practice

For Exercise #5, you will write a program that does the following:

Part A: Write a program that aligns all the following words to the right as shown in the example output:

Dog, Cat, Alligator, Velociraptor, Ox, Horse, and German Shepard

```
***          Dog ***
***          Cat ***
***    Alligator ***
*** Velociraptor ***
***          Ox ***
***          Horse ***
*** German Shepard ***
```

Exercise #5 Part A: Example Output

Part B: Revisiting Exercise 4, adjust the output so that it has the unedited number of miles in addition to a rounded (to two decimal places) as shown in the output.

```
Please enter in a quantity of cookies: 1892
Your inputed quantity of cookies is equal to 1554.7113821138212 miles
Your inputed quantity of cookies is approximately equal to 1554.71 miles
```

Exercise #5 Part B: Example Output